This series aims to report new developments in mathematical research and teaching - quickly, informally and at a high level. The type of material considered for publication includes:

1. Preliminary drafts of original papers and monographs
2. Lectures on a new field, or presenting a new angle on a classical field
3. Seminar work-outs
4. Reports of meetings

Texts which are out of print but still in demand may also be considered if they fall within these categories.

The timeliness of a manuscript is more important than its form, which may be unfinished or tentative. Thus, in some instances, proofs may be merely outlined and results presented which have been or will later be published elsewhere.

Manuscripts should comprise not less than 100 pages.

Publication of *Lecture Notes* is intended as a service to the international mathematical community, in that a commercial publisher, Springer-Verlag, can offer a wider distribution to documents which would otherwise have a restricted readership. Once published and copyrighted, they can be documented in the scientific literature. -

Manuscripts

Manuscripts are reproduced by a photographic process; they must therefore be typed with extreme care. Symbols not on the typewriter should be inserted by hand in indelible black ink. Corrections to the typescript should be made by sticking the amended text over the old one, or by obliterating errors with white correcting fluid. Authors receive 75 free copies.

The typescript is reduced slightly in size during reproduction; best results will not be obtained unless the text on any one page is kept within the overall limit of 18 x 26.5 cm (7 x 10½ inches). The publishers will be pleased to supply on request special stationery with the typing area outlined.

Manuscripts in English, German or French should be sent to Prof. Dr. A. Dold, Mathematisches Institut der Universität Heidelberg, 69 Heidelberg/Germany, Tiergartenstraße or Prof. Dr. B. Eckmann, Eidgenössische Technische Hochschule, CH-8006 Zürich/Switzerland.

Lecture Notes in Physics

Bisher erschienen/Already published

Vol. 1: J. C. Erdmann, Wärmeleitung in Kristallen, theoretische Grundlagen und fortgeschrittene experimentelle Methoden. II, 283 Seiten. 1969. DM 20,-

Vol. 2: K. Hepp, Théorie de la renormalisation. III, 215 pages. 1969. DM 18,-

Vol. 3: A. Martin, Scattering Theory: Unitarity, Analytic and Crossing. IV, 125 pages. 1969. DM 14,-

Vol. 4: G. Ludwig, Deutung des Begriffs physikalische Theorie und axiomatische Grundlegung der Hilbertraumstruktur der Quantenmechanik durch Hauptsätze des Messens. XI, 469 Seiten.1970. DM 28,-

Vol. 5: M. Schaaf, The Reduction of the Product of Two Irreducible Unitary Representations of the Proper Orthochronous Quantummechanical Poincaré Group. IV, 120 pages. 1970. DM 14,-

Vol. 6: Group Representations in Mathematics and Physics. Edited by V. Bargmann. V, 340 pages. 1970. DM 24,-

Vol. 7: R. Balescu, J. L. Lebowitz, I. Prigogine, P. Résibois, Z. W. Salsburg, Lectures in Statistical Physics. V, 181 pages. 1971. DM 18,-

Vol. 8: Proceedings of the Second International Conference on Numerical Methods in Fluid Dynamics. Edited by M. Holt. IX, 462 pages. 1971. DM 28,-

Lecture Notes in Mathematics

A collection of informal reports and seminars
Edited by A. Dold, Heidelberg and B. Eckmann, Zürich

228

Conference on Applications of Numerical Analysis

Held in Dundee/Scotland, March 23-26, 1971

Edited by
John Ll. Morris, University of Dundee, Dundee/Scotland



Springer-Verlag
Berlin · Heidelberg · New York 1971

AMS Subject Classifications (1970): 05 A 17, 30 A 08, 41 A 20, 45 A 55, 45 A 63, 45 B 05, 45 D 05, 45 E 10, 45 L 10, 47 B 45, 49 C 05, 49 D 10, 49 D 15, 49 D 99, 65 D 15, 65 D 30, 65 F 15, 65 H 05, 65 K 05, 65 L 05, 65 L 10, 65 L 15, 65 L 99, 15 M 05, 65 M 10, 65 N 05, 65 N 10, 65 N 15, 65 N 30, 65 N 99, 65 R 05, 90 C 20, 90 C 30, 90 C 50

ISBN 3-540-05656-4 Springer-Verlag Berlin · Heidelberg · New York
ISBN 0-387-05656-4 Springer-Verlag New York · Heidelberg · Berlin

This work is subject to copyright. All rights are reserved, whether the whole or part of the material is concerned, specifically those of translation, reprinting, re-use of illustrations, broadcasting, reproduction by photocopying machine or similar means, and storage in data banks.

Under § 54 of the German Copyright Law where copies are made for other than private use, a fee is payable to the publisher, the amount of the fee to be determined by agreement with the publisher.

© by Springer-Verlag Berlin · Heidelberg 1971. Library of Congress Catalog Card Number 78-177563. Printed in Germany.

Offsetdruck: Julius Beltz, Hemsbach.

F O R E W O R D

During the four days 23rd - 26th March, 1971 at the University of Dundee, Scotland, one hundred and seventy participants attended a conference on the Applications of Numerical Analysis. As the title suggests, the conference was intended to give an opportunity to researchers to present papers and hear the results of others' investigations in Numerical Analysis where methods and analyses were applied to particular problems or with some particular problem in mind. The theme of the conference grew out of the realization that Numerical Analysts, on the one side, and those research workers who meet real life physical problems, on the other side, have been in danger for some time of losing the ability to communicate with one another. We are only too familiar with the numerical analyst who produces a new algorithm and tests the method on a simple (well behaved) problem and claims, as the result of one or two tests, a fabulous new scheme which will solve all practical problems covered by the class of equations for which the scheme is designed. Similarly one is also only too familiar with the engineer (say) who when confronted with the reason for using a particular method states that he has discovered this method in a text book long out of date and uses the method therein simply because he found the notation easy to understand. Often as not the method does not work for the particular problem he has in mind or, perhaps there exist far better methods discovered over the last few years.

Are we to blame the numerical analyst and engineer? If the reason for the former not testing his algorithm rigourously is complacency then clearly the numerical analyst is at fault and he is consequently not producing what ought to be produced; i.e. a good working algorithm with a precise indication of the limitations on the method so that a non-specialist can see at a glance whether or not such a method is capable of solving his problem and in this context some indication of the new algorithm's merits relative to existing algorithms in the field. If, through bad communications with the people who have the problems, the numerical analyst is unable to gain access to real life problems then it is surely the system of communication which is at fault. This problem brings to mind the many papers one reads where the author either does not mention or glosses over those important auxiliary data (for example initial and boundary conditions for a physical problem involving partial differential equations) which make a solution possible or impossible.

Likewise, if our engineer does not use the best methods available simply because it is too much trouble for him to search out the requisite algorithm then there is little one can do to help him. However, as I think occurs in the majority of cases, the engineer is unable to understand what the numerical analyst is trying to say and, what is as bad, is often unable to gain access to the very man who could help him. In this case, once again, it is surely the system of communication which needs review.

The reader may be quick to point out that some groups do exist (for example the Oxford group) which endeavour to produce the necessary links between numerical analysts and 'those with the problems'. However, unfortunately, these instances are in the minority at present. Thus it is hoped that this conference (and others like it) will contribute something towards closing the communication gap. It was therefore gratifying to find at the conference many participants from establishments which work on the physical problems alluded to above. Furthermore, the British Theoretical Mechanics Colloquium (BTMC) was held in Dundee the week after the Applications of Numerical Analysis conference and it was hoped that these 'men of Applied Mathematics' would avail themselves of a numerical analyst colloquium directed towards the BTMC-like problems. Conversly it was hoped that participants at the first conference would attend some of the lectures of the second one. To a certain extent this was true but there is always room for improvement. Thus hopefully conferences such as the subject one of these proceedings will afford both sides the opportunity of communicating with one another and allow the workers on both sides to becomes more familiar with the other's point of view.

The conference took the form of eighteen one-hour lectures given by the invited speakers followed intermittently by half-hour lectures given by the authors of submitted papers, some of these latter talks being presented in parallel sessions. The full list of contributors follows this foreword. Because the theme of the conference was Applications rather than a particular branch of Numerical Analysis, the papers contained in the Volume cover a considerable breadth of interests. For example, function approximation, integral equations, ordinary and partial differential equations of inital and boundary value types, eigenvalue and inverse eigenvalue problems, optimization, and solution of polynomial equations.

As far as possible, we have endeavoured to secure all the papers given at the conference for inclusion in this Volume. However, unfortunately, five of the invited papers have not been included. These are the papers by Fox, Morton, Wilkinson, Golub and Powell. The latter two authors have produces short summaries which are included in the proceedings. The references to the full texts of the talks presented at the conference can be found at the end of this foreword.

The present Volume by Springer Verlag represents the second proceedings of conferences held at Dundee during the Numerical Analysis Year. This project has been financed by the Science Research Council and we acknowledge with gratitude their generous support. It is true to say that the academic year 1970/71 has been extremely exciting in Dundee and to a great extent this is due to the generous support of the Science Research Council. At the time of writing this foreword, a conference on Nonlinear Optimization has just finished when over two hundred participants attended a four day conference. The proceedings of this conference will be published by Academic Press, edited by F. Lootsma. During the next week another conference is to take place on the Ritz-Galerkin procedures and the Finite Element Method when other S.R.C. supported speakers will visit Dundee to present papers, namely Birkhoff, Douglas, Strang, Schoenberg, Thomée and Wachspress. The proceedings of this meeting will be the third this year to be published in the Springer Verlag series. This will be edited by R. Wait.

A considerable amount of the typing in the Volume has been undertaken by Yvonne Nédélec, in the Department of Mathematics. This work is gratefully acknowledged.

Finally, I wish to acknowledge the profound influence that Professor A. R. Mitchell has exercised over this and other proceedings during the Year. The success of the Numerical Analysis Year can be attributed largely to his organisation and guidance.

Dundee, July 1971 John Ll. Morris

References

Fox,L., Mayers, D.F., Taylor, A.B. and Ockendon, J.R.
The Numerical Analysis of a Functional Differential Equation.
Submitted to J.I.M.A., 1971.

Lewis, H.R., Morton, K.W., Sykes, A. and Wesson, J.H.
Comparison of some particle-in-cell plasma simulation methods.
To appear in J. Comp. Phys., 1971.

Wilkinson, J.H. Theoretical and practical properties of inverse interation. To appear in proceedings of National Institute of Rome 'Problems in Numerical Analysis', January 1972.

CONTENTS

Invited Papers

R.BELLMAN: On the Identification of Systems and Some Related Questions 1

R.BELLMAN: Invariant Imbedding: Semigroups in Time, Space and Structure 9

J.CEA, R.GLOWINSKI et J.C. NEDELEC: Minimisation de Fonctionnelles Non-Differentiables 19

L.COLLATZ: Nonlinear Integral Equations and Field Approximation Problems 39

G.H.GOLUB: Some Modified Eigenvalue Problems 56

R.GORENFLO: On Difference Schemes for Parabolic Differential Equations with Derivative Boundary Conditions 57

J.GREENSTADT: Cell Discretization 70

P.HENRICI: Circular Arithmetic and the Determination of Polynomial Zeros 86

R.HUSS and R.KALABA: Computation of the Moments of Solutions of Certain Random Two Point Boundary Value Problems 93

P.LASCAUX et P.A.RAVIART: Stabilité et Précision des Schemas DSN pour l'Equation de Transport en Géométrie Sphérique 103

B.NOBLE: Some Applications of the Numerical Solution of Integral Equations to Boundary Value Problems 137

M.R.OSBORNE: On the Inverse Eigenvalue Problem for Matrices and Related Problems for Difference and Differential Equations. 155

M.J.D.POWELL, I.BARRODALE and F.D.K.ROBERTS: The Differential Correction Algorithm for Rational L_∞ Approximation 169

P.A.RAVIART: Résolution Numérique de Certains Problèmes Hyperboliques Non Linéaires. Méthode de Pseudo-Viscosité 170

H.J.STETTER: Stability of Discretizations on Infinite Intervals 207

E.L.WACHSPRESS: A Rational Basis for Function Approximation . . 223

O.B.WIDLUND: Some Results on Best Possible Error Bounds for Finite Element Methods and Approximation with Piecewise Polynomial Functions 253

Submitted Papers

L.S.CARETTO, A.D.GOSMAN and D.B.SPALDING: Removal of an Instability in a Free Convection Problem 264

G.J.COOPER: Bounds for the Error in Approximate Solutions of Ordinary Differential Equations 270

D.J.EVANS: Numerical Solution of the Sturm Liouville Problem with Periodic Boundary Conditions 277

A.R.GOURLAY, G.McGUIRE and J.Ll.MORRIS: One Dimensional Methods for the Numerical Solution of Nonlinear Hyperbolic Systems . 290

A.JENNINGS: The Development and Application of Simultaneous Iteration for Eigenvalue Problems 297

J.LE FOLL: An Iterative Procedure for the Solution of Linear and Nonlinear Equations 310

J.J.H.MILLER: On Weak Stability, Stability, and the Type of a Polynomial 316

G.M.PHILLIPS: Error Estimates for Certain Integration Rules on the Triangle 321

S.SIGURDSSON: Linear Multistep Methods with Variable Matrix Coefficients 327

J.C.TAYLOR and J.V.TAYLOR: PARODE: A New Representational Method for the Numerical Solution of Partial Differential Equations 332

J.H.VERNER: On Deriving Explicit Runge-Kutta Methods 340

R.WAIT: A Finite Element for Three Dimensional Function Approximation 348

J.R.WHITEMAN, N.PAPAMICHAEL and Q.MARTIN: Conformal Transformation Methods for the Numerical Solution of Harmonic Mixed Boundary Value Problems 353

Papers presented whose proceedings do not appear here:

L.FOX: Numerical Analysis of a Functional Differential Equation

K.W.MORTON: Analysis of Particle Methods for Plasma Simulation

J.H.WILKINSON: Concerning Inverse Iteration

LIST OF SPEAKERS

Invited Speakers

Bellman, R.	Department of Mathematics, Electrical Engineering and Medicine, University of Southern California, Los Angeles, California 90007, U.S.A.
Cea, J.	Université de Nice, UERMST, Nice, France.
Collatz, L.	Institut für Angewandte Mathematik, Universität Hamburg, 2 Hamburg 13, Rothenbaumchaussee 67/69, W. Germany.
Fox, L.	Computing Laboratory, University of Oxford, 19 Parks Road, Oxford, England.
Golub, G.H.	Department of Mathematics, Imperial College, 52/53 Princes Gate, Exhibition Road, London. S.W.7.
Gorenflo, G.	Technische Hochschule, D-51 Aachen, W. Germany.
Greenstadt, J.	I.B.M. Scientific Center, 2670 Hanover Street, Palo Alto, California 94306. U.S.A.
Henrici, P.	Eidgenössische Technische Hochschule, Zürich Switzerland.
Kalaba, R.	Biomedical Engineering, Graduate Center for Engineering Sciences, University of Southern California, University Park, Los Angeles, California 90007. U.S.A.
Lascaux, P.	Commissariat à l'Energie Atomique, Paris, France.
Morton, K.W.	Culham Laboratory, Abingdon, Berkshire, England.
Noble, B.	Mathematics Research Center, University of Wisconsin, Madison 53706. U.S.A.
Osborne, M.R.	Computer Center, Australian National University, Canberra, N.S.W., Australia.
Powell, M.J.D.	Mathematics Branch, A.E.R.E., Harwell, England.
Raviart, P.	Université de Paris, Paris VI, France.
Stetter, H.J.	Technical University of Vienna, Vienna, Austria.
Wachspress, E.L.	General Electric Company, Schenectady, New York, U.S.A.
Widlund, O.B.	Courant Institute of Mathematical Sciences, AEC Computing and Applied Mathematics Center, New York University, 251 Mercer Street, New York, N.Y. 10012, U.S.A.
Wilkinson, J.	N.P.L., Teddington, Middlesex, England.

Other Speakers (submitted papers)

Caretto, L.S.	Mechanical Engineering Department, Imperial College of Science and Technology, London, S.W.7.
Cooper, G.T.	School of Mathematics and Physical Sciences, University of Sussex, Brighton. BN1 9QH.
Evans, D.J.	Computing Laboratory, Department of Applied Mathematics and Computing Science, University of Sheffield, Sheffield 10.
Jennings, A.	Civil Engineering Department, Queen's University, Belfast, N. Ireland.
Le Foll, J.	Research and Development Laboratories, C.A. Parsons and Company Ltd., Newcastle-upon-Tyne. NE6 2YL.
Miller, J.J.H.	School of Mathematics, Trinity College, Dublin, Ireland.
Morris, J.Ll.	Department of Mathematics, University of Dundee, Dundee, Scotland.
Phillips, G.M.	Department of Applied Mathematics, University of St. Andrews, St. Andrews, Scotland.
Sigurdsson, S.	Department of Mathematics, University of Dundee, Dundee, Scotland.
Taylor, J.C.	Department of Natural Philosophy, University of Glasgow, Glasgow, Scotland.
Verner, J.H.	Department of Mathematics, Queen's University, Kingston, Ontario, Canada.
Wait, R.	Department of Mathematics, University of Dundee, Dundee, Scotland.
Whiteman, J.R.	Department of Mathematics, Brunel University, Kingston Lane, Uxbridge, Middlesex, England.